HW 2 posted today or tomorrow, due 9/17

Recall:

- If $u \in C^2(U)$ and $\Delta u = 0$,

  then $\boxed{u(x) = \fint_{\partial B(x,r)} u \, dS = \fint_{B(x,r)} u \, dy}$

  $\boxed{\text{for any ball } B(x,r) \text{ with } \overline{B(x,r)} \subset U}$

  Mean Value Property

- Conversely, if $u \in C^2(U)$ and satisfies $\boxed{\text{Mean Value Property}}$, then $\Delta u = 0$.

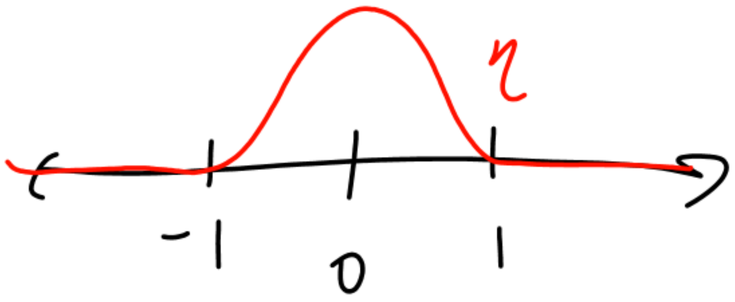
<u>Def</u> u is $C^\infty(U)$ if all partial derivatives of u, of all orders, exist.

Goal: Harmonic functions are C^∞.

Tool: Mollification

Let η be a C^∞ function on $\mathbb{R}^n$ with $\eta \geq 0$, $\int_{\mathbb{R}^n} \eta \, dx = 1$, and $\eta = 0$ outside $B(0,1)$.

$$\eta(x) = \begin{cases} \exp\left(\frac{1}{|x|^2-1}\right) C, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$$



Given a function u, define

$$u_\varepsilon(x) = u * \eta_\varepsilon, \qquad \eta_\varepsilon(x) = \varepsilon^{-n}\eta\left(\frac{x}{\varepsilon}\right)$$

$$= \int_{\mathbb{R}^n} u(x-y)\,\eta_\varepsilon(y)\,dy$$

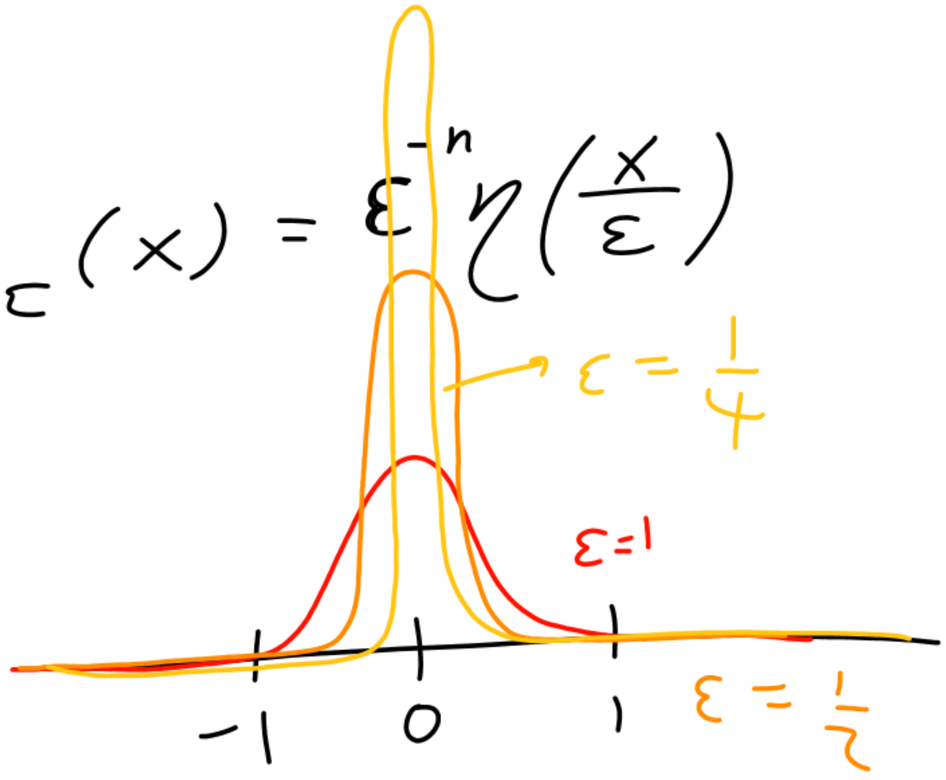
$$= \int_{\mathbb{R}^n} u(x-y)\,\varepsilon^{-n}\eta\left(\frac{y}{\varepsilon}\right)dy$$

$$z = \frac{y}{\varepsilon}$$

$$dz = \frac{dy}{\varepsilon^n}$$

$$= \int_{\mathbb{R}^n} u(x-\varepsilon z)\,\eta(z)\,dz$$

$$\xrightarrow[\varepsilon \to 0]{} \int_{\mathbb{R}^n} u(x)\,\eta(z)\,dz = u(x)\int_{\mathbb{R}^n}\eta(z)\,dz = u(x)$$



We've shown $u_\varepsilon(x) \to u(x)$ pointwise.

Also, u_ε is C^∞.

e.g. $$\frac{\partial}{\partial x_i} u_\varepsilon = \frac{\partial}{\partial x_i}\left(u * \eta_\varepsilon\right) = u * \left(\frac{\partial \eta_\varepsilon}{\partial x_i}\right)$$

$\Rightarrow$ u_ε is differentiable.

Same argument $\Rightarrow$ any partial derivative of u_ε exists $\Rightarrow$ u_ε is C^∞.

Also, $\|u_\varepsilon\|_{L^p(\mathbb{R}^n)} \longrightarrow \|u\|_{L^p(\mathbb{R}^n)}$ for all $1 \le p < \infty$.

Thm If $u \in C(U)$ satisfies the mean value property, then it is C^∞.

Proof Let η be standard mollifier, and $u_\varepsilon = u * \eta_\varepsilon$ for some small $\varepsilon > 0$.

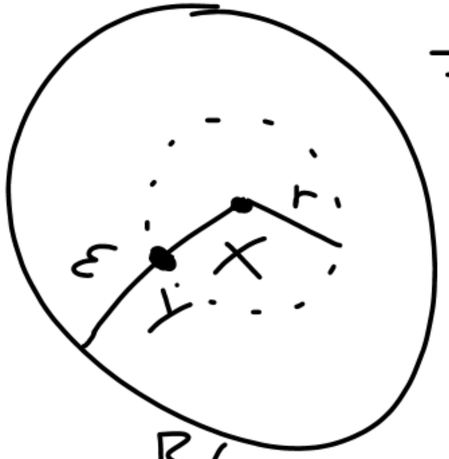
$$u_\varepsilon(x) = \int_{\mathbb{R}^n} u(x-y) \frac{1}{\varepsilon^n} \eta\left(\frac{y}{\varepsilon}\right) dy$$

$$= \int_{B(x,\varepsilon)} u(y) \frac{1}{\varepsilon^n} \eta\left(\frac{|x-y|}{\varepsilon}\right) dy$$

$\eta(z) \neq 0$ if $|z| < 1$

$\eta\left(\frac{x-y}{\varepsilon}\right) \neq 0$ if $|x-y| < \varepsilon$.

$$u_\varepsilon(x) = \int_{B(x,\varepsilon)} u(y)\, \eta\left(\frac{|x-y|}{\varepsilon}\right) \frac{dy}{\varepsilon^n}$$



$B(x,\varepsilon)$

$$= \int_0^\varepsilon \int_{\partial B(x,r)} u(y)\, \eta\left(\frac{r}{\varepsilon}\right) \frac{1}{\varepsilon^n}\, d\omega\, dr$$

$r = |y-x|$

Polar words

$y = x + r\omega$

$y \in \partial B(x,r)$

$dy = r^{n-1}\, dr\, d\omega$

$y = x + r\omega$

$\omega \in \partial B(0,1)$

By Mean Value Property,

$$u(x) = \fint_{\partial B(x,r)} u(y)\, dS$$

$$u(x) = \frac{1}{n\alpha(n) r^{n-1}} \boxed{\int_{\partial B(x,r)} u(y)\, d\omega}$$

$$u_\varepsilon(x) = \int_0^\varepsilon \int_{\partial B(x,r)} u(y)\, dS\, \eta\left(\frac{r}{\varepsilon}\right) \frac{1}{\varepsilon^n}\, dr$$

$$= \int_0^\varepsilon u(x)\, n\alpha(n) r^{n-1} \eta\left(\frac{r}{\varepsilon}\right) \frac{1}{\varepsilon^n}\, dr$$

$$= \frac{n\alpha(n)}{\varepsilon^n} u(x) \int_0^\varepsilon r^{n-1} \eta\left(\frac{r}{\varepsilon}\right) dr$$

$$n\alpha(n) r^{n-1} = \int_{\partial B(x,r)} dS$$

$$= u(x) \int_0^\varepsilon \int_{\partial B(x,r)} \eta\left(\frac{r}{\varepsilon}\right) dS \frac{dr}{\varepsilon^n} = \frac{u(x)}{\varepsilon^n} \int_{B(x,\varepsilon)} \eta\left(\frac{|x-y|}{\varepsilon}\right) dy$$

$$z = \frac{x-y}{\varepsilon} \longleftarrow = u(x) \int_{B(0,1)} \eta(|z|)\, dz = u(x)$$

Putting everything together, we conclude

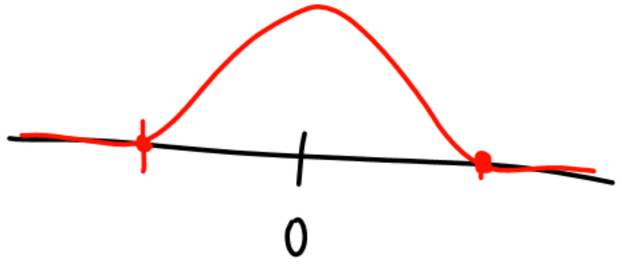
$$\boxed{\text{If } u \in C(U) \text{ and satisfies the Mean Value Property, then } u \in C^\infty \text{ and } \Delta u = 0.}$$

Next, we'd like to show harmonic functions are (real) analytic, i.e.

$$u(x) = \sum_{k=0}^{\infty} \frac{u^{(k)}(x_0)}{k!}(x-x_0)^k \quad (n=1 \text{ case})$$

for every $x_0 \in \mathbb{R}^n$.

$\exp\left(\frac{1}{|x|^2-1}\right)$



To prove analyticity, need <u>quantitative</u> smoothness.

Thm Let u be harmonic in U. Then, for any partial derivative $D^\alpha u$ of order $|\alpha| = k$,

$$|D^\alpha u(x)| \leq C(n,k) \frac{1}{r^{n+k}} \int_{B(x,r)} |u| \, dy$$

$$C(n,k) = \frac{(2^{n+1} n k)^k}{\alpha(n)}, \quad k \geq 1$$

Notation: $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ is a <u>multi-index</u> that tells us how many derivatives to take in each variable.

$$D^\alpha u = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial x_2^{\alpha_2}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} u. \qquad |\alpha| = \alpha_1 + \alpha_2 \cdots + \alpha_n$$

Example

$n = 3, \quad \alpha = (0, 2, 1)$

$$D^\alpha u = \frac{\partial^2}{\partial x_2^2} \frac{\partial}{\partial x_3} u = u_{x_2 x_2 x_3}$$

$$= \frac{\partial^2}{\partial y^2} \frac{\partial}{\partial z} u \qquad |\alpha| = 3$$

Proof of Theorem:

$k=0$: Follows from Mean Value Property

$k=1$: If $\Delta u = 0$, then

$$\Delta u_{x_i} = \frac{\partial}{\partial x_i}(\Delta u) = 0 \Rightarrow u_{x_i} \text{ is also harmonic.}$$

$\Rightarrow u_{x_i}$ satisfies Mean Value Property.

For $r>0$ small enough,

$$u_{x_i}(x) = \fint_{B(x,r/2)} u_{x_i}\,dy = \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\int_{B(x,\frac{r}{2})} u_{x_i}\,dy$$

$$\int_{B(x,r/2)} u_{x_i}\,dy$$

Fact: $u_{x_i} = \operatorname{div}\left[u\vec{e}_i\right]$

where $\vec{e}_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ e.g. $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

Check: $\operatorname{div} F = \sum_{j=1}^{n} (F_j)_{x_j}$

$\operatorname{div}\left[u\vec{e}_i\right] = (u)_{x_i} = u_{x_i}$

Divergence Theorem:

$$\int_{\partial B(x,\frac{r}{2})} F\cdot\vec{\nu}\,dS = \int_{B(x,\frac{r}{2})} (\operatorname{div} F)\,dx$$

$$\int_{B(x,r/2)} u_{x_i}\,dy = \int_{\partial B(x,r/2)} u\vec{e}_i\cdot\vec{\nu}\,dS$$

$$|u_{x_i}(x)| = \frac{1}{\alpha(n)\left(\frac{r}{2}\right)^n}\left|\int_{\partial B(x,r/2)} u\,\vec{e}_i\cdot\vec{\nu}\,dS\right|$$